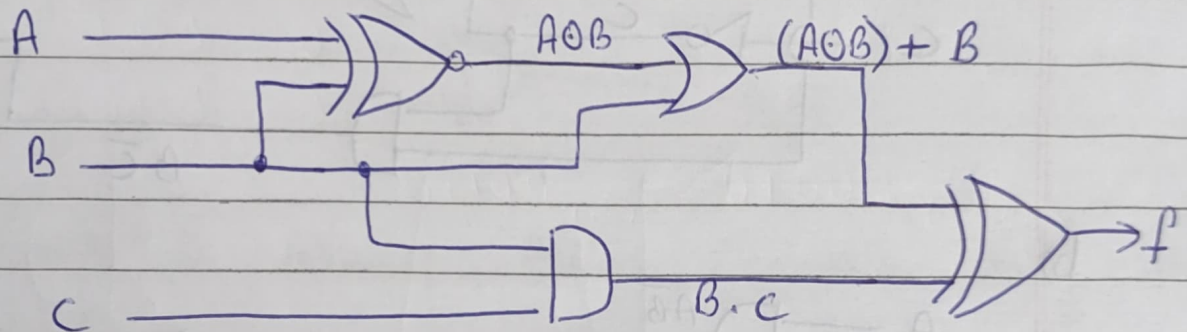


## Day - 2 Boolean Algebra - Assignment

1) Derive the output expression 'f' for the below circuits and draw the simplified circuits

a)



⇒ From circuit;

$$f = ((A \oplus B) + B) \oplus (B \cdot C)$$

Since,  $A \oplus B = AB + \bar{A}\bar{B}$

$$f = (AB + \bar{A}\bar{B} + B) \oplus (B \cdot C)$$

$$= (B(A + 1) + \bar{A}\bar{B}) \oplus (B \cdot C)$$

$$= (B + \bar{A}\bar{B}) \oplus (B \cdot C)$$

$$= ((B + \bar{A})(B + \bar{B})) \oplus (B \cdot C)$$

$$f = (B + \bar{A}) \oplus (B \cdot C)$$

$$\because A \oplus B = \bar{A}B + A\bar{B}$$

$$\Rightarrow f = (B + \bar{A})(\bar{B} \cdot C) + (\bar{B} + A)(B \cdot C)$$

$$= (B + \bar{A})(\bar{B} + C) + (\bar{B} \cdot A)(B \cdot C)$$

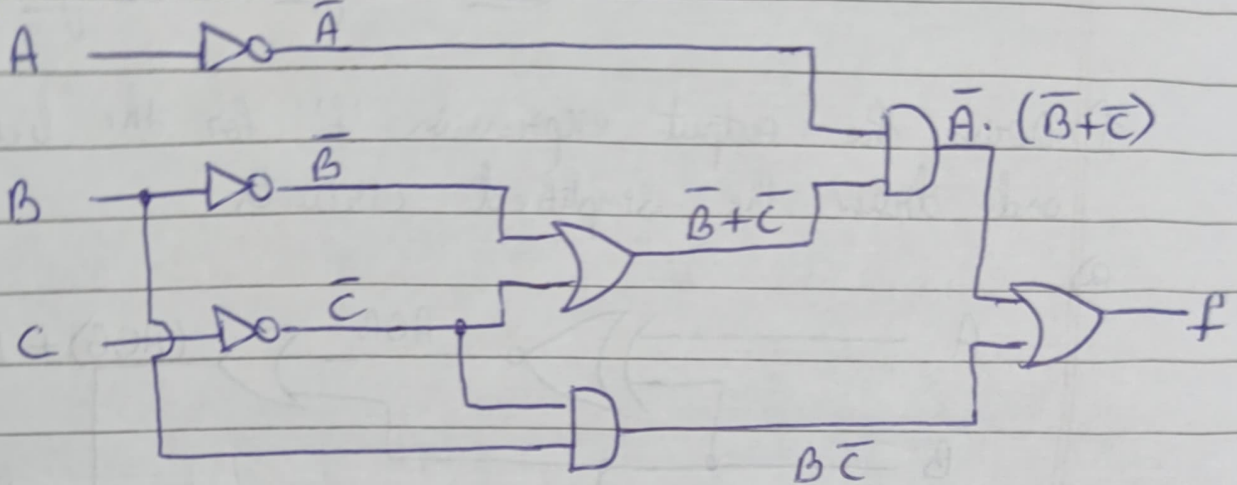
$$= B\bar{B} + B\bar{C} + \bar{A}\bar{B} + \bar{A}C + ((\bar{B} \cdot A)(B \cdot C))$$

$$= B\bar{C} + \bar{A}\bar{B} + \bar{A}C + (\bar{B} \cdot A \cdot B \cdot C) \quad (\because B \cdot \bar{B} = 0)$$

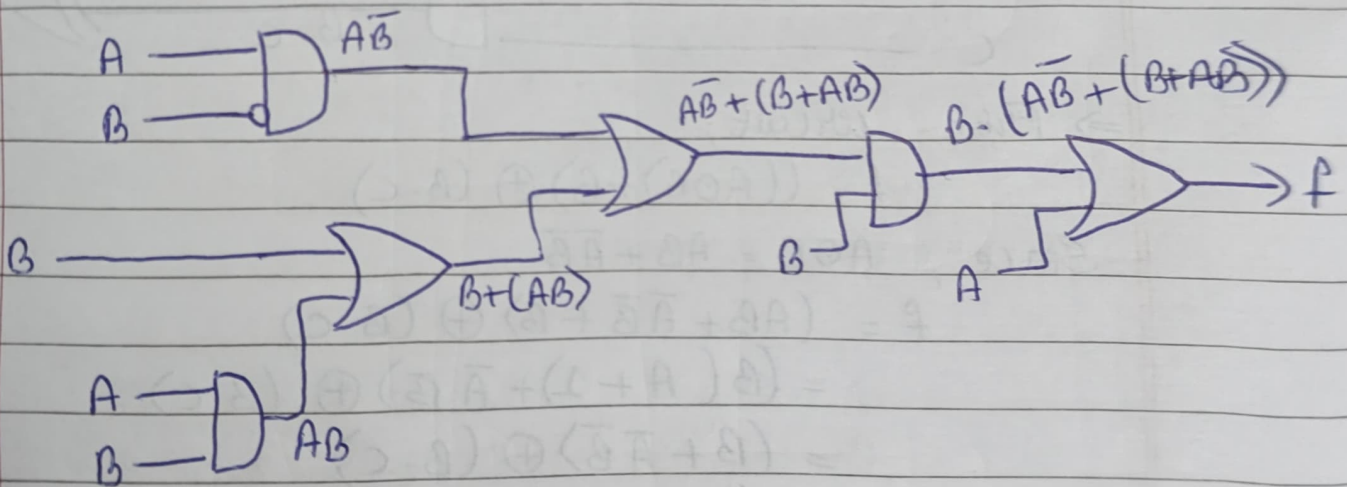
$$\therefore f = \bar{A}\bar{B} + \bar{A}C + B\bar{C}$$

$$f = \bar{A}(\bar{B} + \bar{C}) + B\bar{C}$$

Simplified Circuit



b)



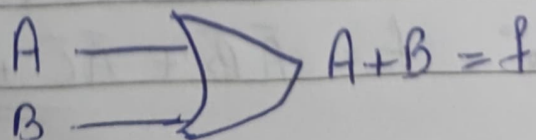
$$\Rightarrow f = A + (B \cdot (\bar{A}\bar{B} + (B + AB)))$$

$$f = A + (B \cdot (\bar{A}\bar{B} + B))$$

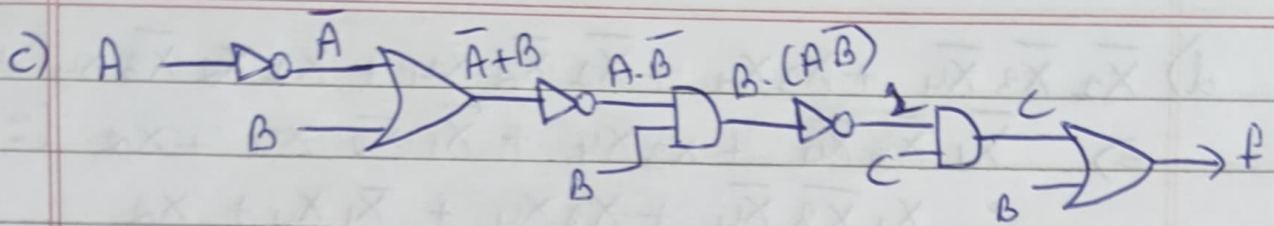
$$f = A + (B \cdot B)$$

$$\therefore \underline{\underline{f = A + B}}$$

Simplified Circuit;





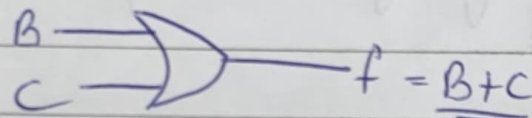


$$\Rightarrow F = B + C$$

$$\overline{A+B} = A \cdot B$$

$$B = A \cdot B$$

Simplified Circuit



2) Minimize the following Boolean Expression using Boolean postulates.

a)  $\overline{A}C\overline{D}E + \overline{A}\overline{B}\overline{D} + ABCE + ABD$

$$\Rightarrow \overline{A}C\overline{D}E + \overline{A}\overline{B}\overline{D} + AB(CE + D)$$

$$\therefore \overline{A}\overline{D}(CE + \overline{B}) + AB(CE + D)$$

==

b)  $\overline{A}\overline{B}\overline{C} + ABD + \overline{A}C + \overline{A}C\overline{D} + A\overline{C}\overline{D} + A\overline{B}\overline{C}$

$$\Rightarrow \overline{B}\overline{C}(\overline{A} + A) + ABD + \overline{A}C + \overline{A}C\overline{D} + A\overline{C}\overline{D}$$

$$(\because A + \overline{A} = 1)$$

$$\overline{B}\overline{C} + ABD + \overline{A}C + \overline{A}C\overline{D} + A\overline{C}\overline{D}$$

$$\overline{B}\overline{C} + ABD + \overline{A}C(1 + \overline{D}) + A\overline{C}\overline{D}$$

$$\overline{B}\overline{C} + ABD + \overline{A}C + A\overline{C}\overline{D}$$

==

c)  $\overline{A}\overline{C} + BC + A\overline{B} + \overline{A}BC + AC\overline{D} + \overline{B}\overline{C}\overline{D}$

$$\Rightarrow \overline{A}\overline{C} + BC(1 + \overline{A}) + A\overline{B} + AC\overline{D} + \overline{B}\overline{C}\overline{D}$$

$$\overline{A}\overline{C} + BC + A\overline{B} + AC\overline{D} + \overline{B}\overline{C}\overline{D}$$

==

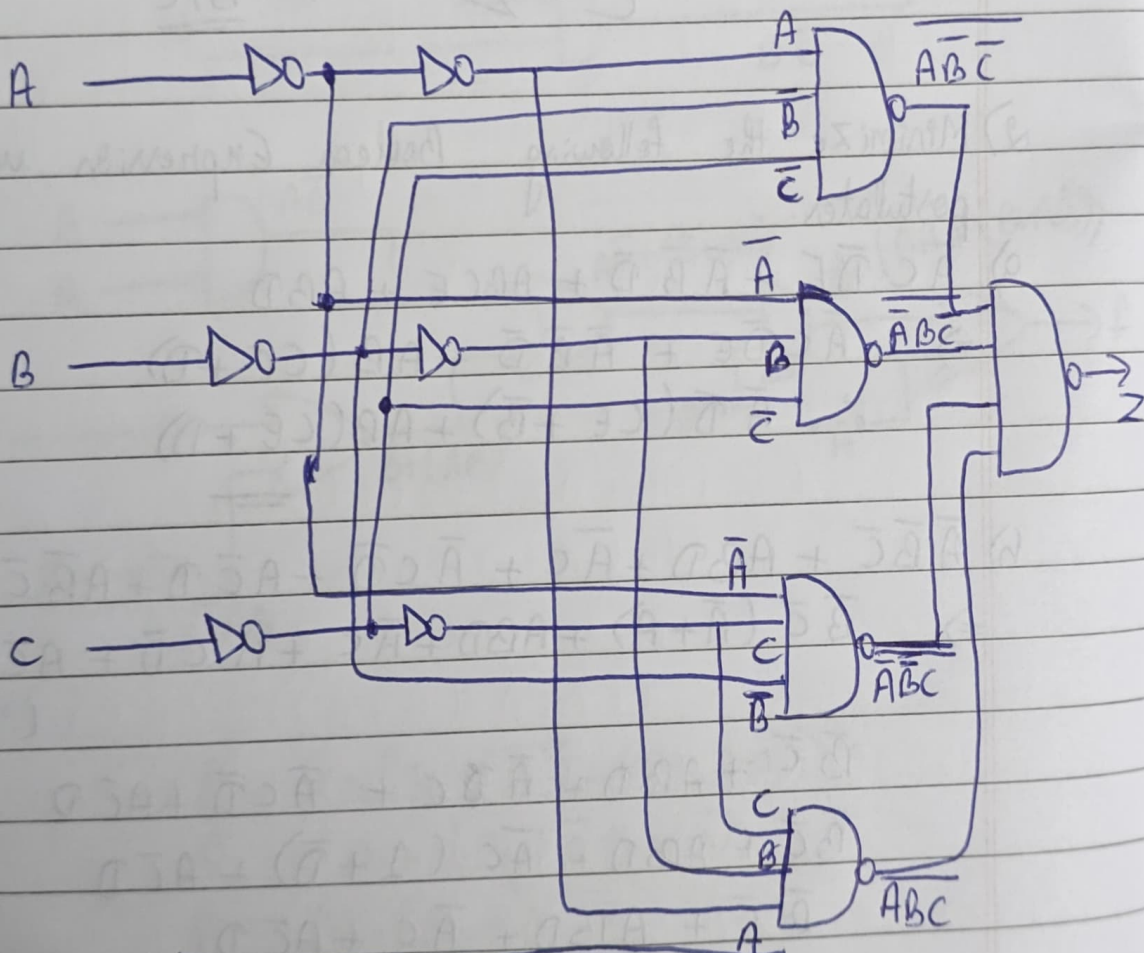
$$d) \bar{X}_1 \bar{X}_3 \bar{X}_4 + X_3 X_4 + \bar{X}_1 \bar{X}_2 X_4 + X_1 X_2 \bar{X}_3 X_4 + X_2$$

$$\Rightarrow \bar{X}_1 \bar{X}_3 \bar{X}_4 + X_3 X_4 + \bar{X}_1 \bar{X}_2 X_4 + X_2 \quad (\because 1+A=1)$$

$$\therefore X_1 \bar{X}_3 \bar{X}_4 + X_3 X_4 + \bar{X}_1 X_4 + X_2$$

$$(\because X + \bar{X}Y = \bar{X} + Y)$$

3) Which single gate can represent the functionality of given circuit?



$$\Rightarrow Z = (\bar{A}\bar{B}\bar{C})(\bar{A}B\bar{C})(\bar{A}B\bar{C})(\bar{A}B\bar{C})$$

$$= (\bar{A}\bar{B}\bar{C}) + (\bar{A}B\bar{C}) + (\bar{A}B\bar{C}) + (\bar{A}B\bar{C})$$

$$= \bar{C}(\bar{A}\bar{B} + \bar{A}B) + C(\bar{A}B + \bar{A}B)$$

$$= \bar{C}(A \oplus B) + C(A \oplus B)$$



$$\therefore \overline{A \oplus B} = A \odot B$$

$$\Rightarrow Z = \overline{A \oplus B} + C(\overline{A \oplus B})$$

$$Z = (A \oplus B) \oplus C$$

$$\therefore Z = A \oplus B \oplus C$$

$\therefore$  The single gate representing the circuit is a 3 input XOR gate.

4) Minimize the following Boolean Expressions using K-Map

a)  $f(a, b, c, d) = \sum m(4, 6, 8, 10, 11, 12, 15) + \sum d(3, 5, 7, 9)$

| ab \ cd          | $\bar{c}\bar{d}$ | $\bar{c}d$ | $c\bar{d}$ | $cd$ |
|------------------|------------------|------------|------------|------|
| $\bar{a}\bar{b}$ | 0                | 0          | X          | 0    |
| $\bar{a}b$       | 1                | X          | X          | 1    |
| $a\bar{b}$       | 1                | 0          | 1          | 0    |
| $ab$             | 1                | X          | 1          | 1    |

$$f = \bar{a}b + a\bar{b} + cd + a\bar{c}\bar{d}$$

$$\therefore f = (a \oplus b) + \underline{cd} + \underline{a\bar{c}\bar{d}}$$

b)  $f(a, b, c, d) = \Pi_m(0, 2, 4, 6, 7, 9) \cdot \Pi_d(10, 11)$

| a+b \ c+d         | $c+d$ | $c+\bar{d}$ | $\bar{c}+\bar{d}$ | $\bar{c}+d$ |
|-------------------|-------|-------------|-------------------|-------------|
| $a+b$             | 0     | 1           | 1                 | 0           |
| $a+\bar{b}$       | 0     | 1           | 0                 | 0           |
| $\bar{a}+\bar{b}$ | 1     | 1           | 1                 | 1           |
| $\bar{a}+b$       | 1     | 0           | X                 | X           |

$$\therefore f = (a+d) \cdot (a+\bar{b}+\bar{c}) \cdot (\bar{a}+b+d)$$